

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT Plan the learning first; let the checklist follow. Click a field to type • click a box to tick it.

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 14 (6-10 Dec 2026) - Period 1	Unit 4 - 4.4 Animations (1/2)
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 14 / Lesson 1 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Games with sound uploaded by ~80%; mute control done by 60%. Today: CSS hover effects and @keyframes animations.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students add hover effects (transition, transform) to buttons and gloves and create CSS @keyframes animations with timing functions. Sprint 4.4 obj. 1-2; Challenges 2-3; CSTA 3A-AP-16.

Success criteria — what will successful learning look / sound like?

I add :hover transitions to buttons and gloves; I write an @keyframes animation with at least 3 stops; I control duration, timing and iteration.

Key knowledge / skill / understanding:

transition animates between 2 states; @keyframes defines stages in %; animation shorthand.

Likely misconception / access or challenge need:

Animations need JavaScript; % in keyframes means size.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Modelling / worked exam ☒ Time: 10 min	Strategy: Experiment / lab ☒ Time: 20 min	Strategy: Gamification ☒ Time: 7 min
What students will do: Code along: hover on buttons (transform: scale, transition) and gloves (rotate), then Activity 1 rotate-scale circle with @keyframes.	What students will do: Animation lab: Challenge 2 floating astronaut with shadow, or Challenge 3 rectangle-to-circle with border-radius; record what changing duration/timing does.	What students will do: 'Guess the animation': teacher shows animations; teams write the keyframe percentages and properties; points for accuracy.
Evidence tool: Mini-whiteboards ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Portfolio / notebook work ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group	Evidence tool: Digital quiz (Forms / Kahoot / Quizizz) ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☒ Group
Resources: VS Code; QR 'CSS animations'; Activity 1 source.	Resources: Challenge pages; lab record sheet.	Resources: Kahoot/Quizizz with animated GIFs.
Support (low performers / SEN): Copy-ready snippet; change values only.	Support (low performers / SEN): Choose Challenge 3 (simpler) with guide.	Support (low performers / SEN): Multiple-choice version.
Stretch (fast learners / gifted): Use cubic-bezier timing.	Stretch (fast learners / gifted): Challenge 1: three vehicles moving in a square path one after the other.	Stretch (fast learners / gifted): Write full @keyframes code for the bonus question.

3 PREPARE — What do I need to do so students can succeed?				4 RESPOND — What will I do with what I see and hear?			
Set students up to succeed: ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources ☒ Praise, recognition and reinforcement planned				If students struggle or finish early I will: ☒ Re-teach ☒ Regroup ☒ Give extra support ☒ Stretch ☒ Have students redo / improve			
My setup / modelling: Time: 6 min Hook: our game vs the same game with bouncing buttons and wobbling gloves. Model transition vs animation; write @keyframes with 0%, 50%, 100%; show animation shorthand and DevTools Animations panel. Success criteria shared.				What I will watch for / likely responses: Animation name mismatch; forgetting duration so nothing moves; too many animations distracting players; motion sensitivity (mention prefers-reduced-motion). Response: 'nothing animates'.			
What is new or innovative in this lesson? <i>(ETEC looks for this)</i> Lab-style experimenting with animation variables and recording results; inclusive design via reduced-motion setting for sensitive users.				Close / exit check <i>(summative + time for learners' questions)</i> Time: 7 min Exit ticket: write a 3-stop keyframe that fades and grows an element. Learners' questions.			
Progress recorded / shared: ☒ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio				Homework / platform <i>(e.g. Madrasati)</i> Madrasati: complete one animation challenge; upload short screen recording. Plan your game's final polish list.			

5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>		
What worked — and how do I know? Animation lab was highly engaging; students compared results scientifically. Kahoot showed 80% understood keyframe percentages.	What will I change next time? Introduce prefers-reduced-motion earlier. Limit lab choice to 2 challenges to save time.	Students needing follow-up: Students behind on game logic (approx. 3 per class) to receive the fixed source to join the final showcase.

COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i>							
Literacy read + write · 2-1-1-6 ☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Numeracy numbers / data · 2-1-1-6 ☒ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Digital students use it · 2-1-1-4 ☒ 1 ☒ 2 ☐ 3 ☐ C ☒ H	Digital research trusted sources · 2-1-1-4 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	Real-life ① application · 2-1-1-5 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	Real-life ② application · 2-1-1-5 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Inquiry experiment / research · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Thinking problem-solving · 2-1-1-7 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H
Create something new · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Discussion expressing ideas · 2-1-1-8 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Pair / group cooperative work · 2-1-1-8 ☐ 1 ☒ 2 ☒ 3 ☐ C ☐ H	Learner choice independent work · 3-2-1-6 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Self-assessment 2-2-1-1 ☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Peer assessment 2-2-1-1 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	SEN / low specific support · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H	Gifted specific stretch · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback
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